Derivation of the Normal Equation (W)

1. The Model and Homogeneous Form

We start with the prediction for a single student i using n features:

$$\hat{y}_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_n x_{in}$$

To simplify, we use **Homogeneous Form** by adding a dummy feature $x_0 = 1$. This lets us treat the intercept β_0 as just another weight w_0.

- **Weight Vector (W):** $[w_0, w_1, \ldots, w_n]^T$
- **Input Matrix (X):** An $m \times (n+1)$ matrix where the first column is all ones.
- **Prediction Vector:** $\hat{Y} = XW$

2. The Cost Function (SSE)

We define the error as the difference between actual GPA (Y) and predicted GPA ($\hat{Y}$). To find the "best" weights, we minimize the **Sum of Squared Errors (SSE)**:

$$SSE = \sum (y_i - \hat{y}_i)^2 = (Y - XW)^T (Y - XW)$$

3. Matrix Expansion

Expanding the quadratic expression:

$$SSE = Y^T Y - Y^T XW - (XW)^T Y + (XW)^T XW$$

Since $Y^T XW$ is a scalar, it equals its transpose $W^T X^T Y$. We combine them:

$$SSE = Y^T Y - 2W^T X^T Y + W^T X^T XW$$

4. Differentiation & The Normal Equation

To find the minimum, we take the derivative with respect to W and set it to zero:

$$\frac{\partial SSE}{\partial W} = -2X^T Y + 2X^T XW = 0$$

Dividing by 2 and rearranging gives us the **Normal Equation**:

$$(X^T X)W = X^T Y$$

5. Final Goal

Multiplying by the inverse of the first term:

$$W = (X^T X)^{-1} X^T Y$$

What do these matrices actually represent?

In the final formula $W = G^{-1}M$, we have two critical components:

1. The Gram Matrix ($G = X^T X$)

- **What it is:** A square matrix ($n + 1 \times n + 1$) that contains the dot products of all your input features.
- **What it represents:** It captures the **relationships between your features**. The diagonal values represent the "strength" (variance) of each feature, and the off-diagonal values show how much features "overlap" (covariance).
- **Why it's important:** It tells the model if your features are redundant. If two features are perfectly correlated (e.g., "Hours Studied" and "Minutes Studied"), the Gram Matrix becomes non-invertible, and the math "breaks"—this is called **Multicollinearity**.

2. The Moment Matrix ($M = X^T Y$)

- **What it is:** A vector that contains the dot product of your input features and the actual target (GPA).
- **What it represents:** It captures the **relationship between your features and the output**. It essentially "measures" how much each feature (like Study Hours) pulls the GPA up or down before considering the other features.
- **Why it's important:** It acts as the raw "signal" for the weights. It tells the model the direction and initial magnitude for every β coefficient.

1. The Gram Matrix as "Information"

The Gram Matrix doesn't care about the target (GPA); it only cares about the **Input Features**. It captures how much "signal" is available in your data and how "clean" that signal is.

- **The Diagonal Values:** These represent the **Energy/Variance** of each feature.
- **The Off-Diagonal Values:** These represent the **Redundancy/Overlap** between features.

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2. The Diagonal Values ($i = j$)

The values falling on the diagonal (g_{11}, g_{22}, etc.) are the result of a feature being dotted with itself (e.g., $StudyHours \cdot StudyHours$).

- **What they mean:** They represent the total squared magnitude of that feature.
- **High Value:** Indicates a feature with high variance and a wide range of data points. This provides "strong information" for the model to learn from.
- **Low Value:** Indicates a feature that doesn't change much (low variance). If a value is near zero, the feature provides almost no information to help the model distinguish between different students.

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3. Above and Below the Diagonal ($i \neq j$)

Since the Gram Matrix is **Symmetric** ($X^T X = (X^T X)^T$), the values above the diagonal are mirror images of the values below. These are the "Cross-Products" (e.g., $StudyHours \cdot Attendance$).

Positive Values (High Correlation)

- **Meaning:** The two features move together. When Study Hours increase, Attendance also tends to increase.
- **Impact on Model:** This represents **Redundancy**. If the value is extremely high, the Gram Matrix tells the model: *"You don't need both of these features; they are telling the same story."*

Zero Values (Orthogonality)

- **Meaning:** The two features are perfectly independent. Knowing a student's study hours tells you absolutely nothing about their attendance.
- **Impact on Model:** This is the "Ideal Information." It means each feature provides a 100% unique piece of the puzzle for predicting GPA.

Negative Values (Inverse Correlation)

- **Meaning:** The features move in opposite directions. (e.g., perhaps "Hours Spent Gaming" vs "GPA").
- **Impact on Model:** This is still useful information! It tells the model that these two features provide "opposing signals" that help balance the weight matrix.

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4. Why this matters for the "Weight Matrix" (W)

The reason we **invert** the Gram Matrix (G^{-1}) in our final formula ($W = G^{-1}M$) is to **remove the redundancy** found in those off-diagonal values.

- If the off-diagonal values are high, the inversion process "penalizes" the weights so the model doesn't double-count the impact of correlated features.
- If the Gram Matrix is "Singular" (the off-diagonal values are so high that one feature is a perfect copy of another), the matrix cannot be inverted. Mathematically, the model "crashes" because it cannot decide which feature should get the credit for the prediction.